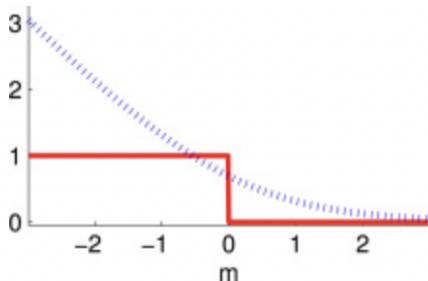


Perceptron

Geometric Interpretation

Procheta Sen



Hyperplane

- ▶ The decision in perceptron is made depending on

$$\overline{W}^T \overline{X} + b > 0 \quad \text{or} \quad \overline{W}^T \overline{X} + b \leq 0$$

- ▶ $\{\overline{X} : \overline{W}^T \overline{X} + b = 0\}$ is the critical region (**decision boundary**)
- ▶ $\overline{W}^T \overline{X} + b = 0$ defines a hyperplane

Example:

- ▶ In 2D space we have $w_1 x_1 + w_2 x_2 = 0$ (ignoring the bias term), which is a straight line through the origin.
- ▶ In N -dimensional space this is an $(N - 1)$ -dimensional hyperplane.

Geometric interpretation

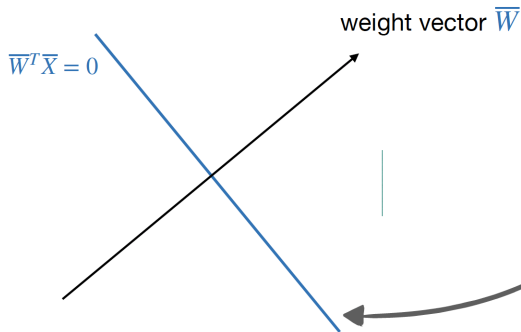


Figure: Hyperplane defined by the weight vector is the line perpendicular to the weight vector.

Geometric Interpretation

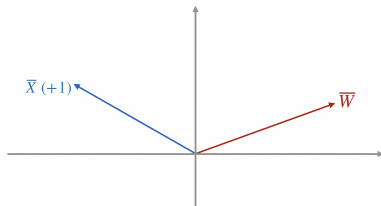


Figure: The angle between the current weight vector $\bar{W}$ and the positive instance $\bar{X}$ is greater than 90° . Therefore, $\bar{W}^T \bar{X} < 0$, and this instance is going to get misclassified as negative.

Geometric Interpretation

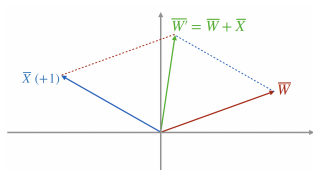


Figure:

- ▶ The new weight vector $\vec{w}'$ is the addition of $\vec{w} + \vec{x}$ (as per the perceptron update rule).
- ▶ It lies in between $\vec{x}$ and $\vec{w}$.
- ▶ Notice that the angle between $\vec{w}'$ and $\vec{x}$ is less than 90° .
- ▶ Therefore, $\vec{x}$ will be classified as positive by $\vec{w}'$.

Vector Algebra Revision

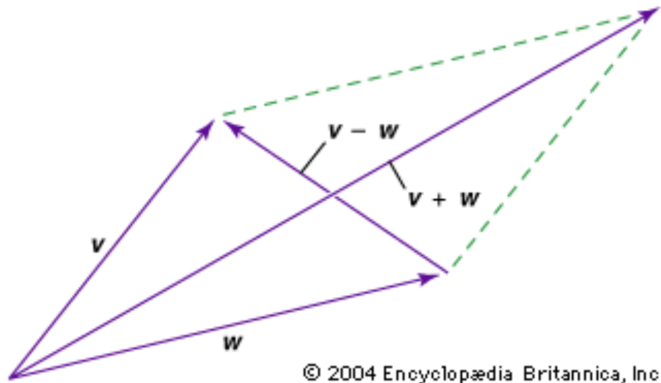


Figure: Vector Algebra Addition

Linear separability

If a given set of positive and negative training instances can be separated into those two groups using a straight line (hyperplane), then we say that the dataset is linearly separable.

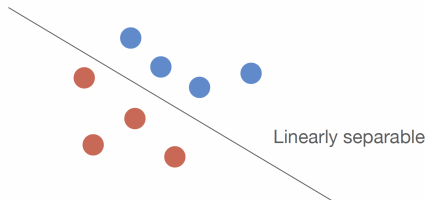


Figure: Vector Algebra Addition

Linear Separability

- ▶ When a dataset is linearly separable, there can exist more than one hyperplanes that separates the dataset into positive/negative groups.
- ▶ In other words, the hyperplane that linearly separates a linearly separable dataset might not be unique.
- ▶ However, (by definition) if a dataset is non-linearly separable, then there exist NO hyperplane that separates the dataset into positive/negative groups.

A non-linearly separable case

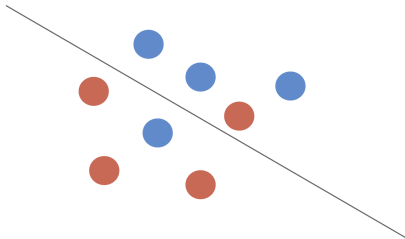


Figure: No matter how we draw straight lines, we cannot separate the red instances from the blue instances

Further Remarks

- ▶ When a dataset is linearly separable it can be proved that the perceptron will always find a separating hyperplane!
- ▶ The final weight vector returned by the Perceptron is more influenced by the final training instances it sees.
- ▶ Take the average over all weight vectors during the training (Averaged Perceptron algorithm)